

Public Seminar

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Enhancing Machine Learning Resilience to Adversarial Manipulations via Moving Target Strategies

by

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Friday December 13 2024, 10:30 AM

Abstract

Machine learning (ML) models are increasingly deployed in high-stakes domains such as autonomous driving, medical diagnostics, financial forecasting, and cybersecurity. However, their vulnerability to adversarial manipulations poses serious security and privacy risks, including training data poisoning, adversarial examples, membership inference, and model stealing. While significant progress has been made in understanding and mitigating these vulnerabilities, many existing defenses remain static, enabling adversaries to repeatedly exploit fixed decision boundaries.

In this talk, I will describe a defense approach that adopts a moving target strategy to enhance model resilience in the face of adversarial examples. By dynamically altering the model's decision boundary, this strategy introduces uncertainty for adversaries, disrupting their ability to craft successful attacks. I will discuss the conceptual underpinnings, practical implementation, and evaluation of this approach, demonstrating its potential to advance trustworthy ML in adversarial environments.

Short Bio

Dr. Birhanu Eshete is an Associate Professor of Computer Science at the University of Michigan-Dearborn, where he leads the Data-Driven Security & Privacy Lab. He is a recipient of the U.S. State Department Fulbright U.S. Scholar Award (2024-2025), the U.S. National Science Foundation CAREER Award (2023), and the USENIX Security Symposium Distinguished Paper Award (2018). His research focuses on trustworthy machine learning, addressing adversarial manipulations, privacy-motivated inference attacks, and the interplay of security, privacy, transparency, and fairness. Dr. Eshete's work has been published in top security & privacy venues such as IEEE S&P, ACM CCS, USENIX Security, ISOC NDSS, the Science Magazine and featured in the NIST Trustworthy AI Resource Center. He holds a Ph.D. from the University of Trento and M.Sc. and B.Sc. degrees from Addis Ababa University.



Venue

Smart Class Room, SECE, AAiT